

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1022

Antrim 138 kV
November 18, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1022. The Interconnection Customer proposes a 152.0 MW wind facility interconnecting at the Antrim 138 kV substation in the Pennsylvania area. The request was received on July 07, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 365 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$30,959,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Otsego – Peterson Flats 138 KV Line	West Waynesboro – Chambersburg 345 KV Line	133.7	347	34.0
Antrim – Peterson Flats 138 KV	West Waynesboro – Chambersburg 345 KV	131.3	366	34.0
Ros Fork – Thurmont 138 KV Line	West Kelso Substation – Dry Gap 138 KV	115.6	202	6.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Otsego Substation – Peterson Flats Substation 138 KV L	Rebuild line with higher-capacity conductor	\$4,284,000	29% (\$1,228,000)
Antrim – Peterson Flats 138 KV	Rebuild line with higher-capacity conductor	\$8,373,000	29% (\$2,400,000)
Ros Fork – Thurmont 138 KV line	Rebuild line with higher-capacity conductor	\$27,331,000	100% (\$27,331,000)
	Total allocated to project		\$30,959,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.